patterns, buying habits, preferences, culture and other characteristics.

IoT for healthcare: IoT based applications/systems of healthcare enhance the traditional technology used today. These devices help increase the accuracy of the medical data collected from a large set of devices connected to various applications and systems. They also help gather data to improve the precision of the medical care delivered through sophisticated integrated healthcare systems.

- IoT devices give direct 24/7 x 365 access to the patient in a less intrusive way. IoT based analytics and automation allow health providers to access patient reports before they arrive at the hospital. They also help improve responsiveness in emergency healthcare.
- IoT-driven systems are used for continuous monitoring of a patient's health status. These employ sensors to collect the physiological information analysed and stored on the cloud. Doctors access this information for further analysis and review. This way, a continuous automated flow of information is provided.
- Patients' health data is captured using various sensors, analysed and sent to the medical professional for proper remote medical assistance.

IoT for education: IoT customises and enhances education by optimising content and forms of delivery.

- *Student tracking:* IoT facilitates the customisation of education to give students access to what they need.
- *Instructor tracking:* Educators can use the IoT platform for customised instructional designs for their students.
- *Facility monitoring and maintenance:* IoT based applications improve the professional development of educators.
- *Data from other facilities:* IoT also enhances the knowledge base used to establish education standards and practices. It introduces large, high-quality, real-world data sets into educational design.

IoT for agriculture: IoT minimises human intervention in farming analysis and monitoring. IoT based systems detect changes in crops, the soil, etc.

- *Crop monitoring:* IoT based devices can be used to monitor crops and the health of plants using the data collected from them. These devices can also monitor pests and diseases proactively in the early stages of infestation.
- *Food safety:* IoT helps connect the entire supply chain for ensuring the food is safe when it reaches the customer.
- *Climate monitoring:* IoT based devices are used to monitor temperature, humidity, the intensity of light, and moisture of the soil. The data is ingested into the central repository to trigger alerts and automate water, air, and crop control.
- *Logistics monitoring:* Location based IoT devices can be used to track vegetables and other farm products during transport and storage.
- *Livestock monitoring:* Farm animals can be monitored using IoT devices to detect potential signs of disease. The data can be analysed from the central repository, and the farmers can share relevant information.

IoT analytics has a number of benefits that are helping transform businesses and different industries.

- Greater transparency; transaction histories are becoming more transparent
- Increased visibility and better control, resulting in quicker decision-making
- Flexible scaling of business needs and expansion in new markets
- Reduction in operational costs due to automation and better resource utilisation
- New revenue streams, resulting from the clearing of operational challenges
- Enhanced customer experience based on an analysis of the purchase history

IoT is one of the most significant sources of Big Data, which is rendered useless without analytics power. IoT analytics comes into play when voluminous amounts of data needs to be processed, transformed, and analysed at a high frequency. It helps to provide advanced services across many domains. END

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